JUST Undergraduate Final Admission System

A Project Report Submitted

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Submitted by

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Preface

On the eve of the beginning of civilization, human beings had been striving tirelessly to discover, explore, and innovate unknown aspects of the world. This inherent curiosity and desire for advancement have driven humanity toward continuous progress and development. Over time, human beings have not only acquired theoretical knowledge but have also emphasized the importance of practical implementation, which plays a crucial role in solving real-world problems and improving the quality of life.

In the modern era, the field of Computer Science and Engineering has emerged as one of the most dynamic and influential disciplines, contributing significantly to technological advancement and digital transformation. The philosophy of the Computer Science and Engineering department is centered on equipping students with both theoretical understanding and practical skills. To achieve this goal, the department has introduced thesis and project programs that encourage students to engage in real-life problem-solving and innovative system development.

As a part of fulfilling the academic requirements of the undergraduate program, I have undertaken a project titled **"JUST Undergraduate Final Admission System"**. This project aims to develop a comprehensive web-based platform that simplifies and streamlines the admission process for undergraduate students. It focuses on efficiency, transparency, and user-friendliness, ensuring a better experience for both applicants and administrators.

Throughout the development of this project, I have conducted extensive research, analysis, and observation to understand the existing admission processes and identify their limitations. This has enabled me to design and implement a system that addresses real-world challenges effectively. The project has provided me with valuable opportunities to enhance my technical skills, problem-solving abilities, and understanding of software development methodologies.

I believe that this project will not only serve as an academic accomplishment but also contribute positively to the automation and digitalization of admission systems. The knowledge and experience gained during this work will undoubtedly play a vital role in my future career and professional development.

Contents

Chapter 1

Introduction

1.1 Background

Admission of a university is an indispensable activity in every educational activity and is as old as education itself. With more students applying for admission than in previous years, admission process for admissions officers has become increasingly time-consuming. More applications resulted in heavier paperwork and processing challenges. Every admission is routed through various faculty for evaluation, and the manual admission process caused difficulty in admission, and archiving.

In order to speed up and simplify the admission process, the development of the Automated Final Admission System is to enact document redesign and an automated management program. The proposed "JUST undergraduate final admission system" would store, route, and retrieve prospective and current application documents. The development of science and technology has immensely contributed to the growth of computers and the internet which has adversely increased the use and need for an automated applicant's admission system.

Universities, Colleges, Companies, organizations, etc. are seeking better ways to distribute and disseminate information throughout the world. In order to achieve this, they develop and rely on online applications, which help them input, update, process, and disseminate information. The internet has increased the use of online applications because data could be easily and readily accessed as well as documented. This project is targeted at allowing admission of applicants' decisions to be made faster and more efficiently. It looks at improving the present traditional system of the manual process (paper-based) which is manually documented. This type of project is a hot topic in the whole world to digitalize our education system.

1.2 Motivation

1.2.1 Existing System

There are several automated systems used for managing final admission processes in educational institutions. These systems differ in features, capabilities, and implementation depending on institutional needs and available resources.

Currently, JUST provides a web-based application titled "JUST Admission 2025". While this system includes some useful functionalities, it is not a fully integrated or complete solution. Several important features are missing, such as admission confirmation, sending admission confirmation messages, and verification of required documents.

Another issue in the current system is that any user can register regardless of whether they have been selected for admission or not. This creates inconsistency and reduces system reliability. Additionally, the system lacks proper validation mechanisms, which makes it difficult to ensure the authenticity and correctness of submitted data.

1.2.2 Limitations of the System

The current system suffers from several limitations that affect its efficiency and usability.

First, there is no proper validation process implemented. As a result, users can input incorrect or incomplete data without restriction. Second, the system allows unrestricted data insertion but does not provide proper mechanisms for updating or removing data, which can lead to data inconsistency.

Third, the system architecture and code structure are not well organized, making it difficult to maintain, scale, or extend the system in the future.

Another significant limitation is that users who are not eligible (i.e., those who did not get a chance for admission) can still register in the system. This reduces the accuracy and trustworthiness of the admission process.

Finally, key features such as admission confirmation, automated message sending, and document verification are either missing or not fully implemented, which prevents the system from functioning as a complete automated admission solution.

1.3 Objectives

The primary objective of this project is to develop a user-friendly and reliable web-based admission system that ensures maximum accuracy and efficiency. The system aims to provide the following facilities:

- **Efficiency:** Automate the admission process to reduce manual effort, save time, and streamline data handling.
- **Accuracy:** Minimize errors by reducing manual data entry and ensuring proper handling of application data and documents.
- **Consistency:** Ensure that all data and applications are processed uniformly, reducing the risk of inconsistency and data loss.
- **Validation and Reliability:** Implement proper validation mechanisms to ensure that only eligible candidates can register and submit accurate information.
- **Improved Performance:** Enhance system performance and reliability by using modern technologies and optimized system architecture.
- **Improved Security:** Strengthen system security by incorporating updated security measures to protect user data and prevent unauthorized access.
- **Cost Effectiveness:** Reduce operational costs by minimizing the need for manual labor and paper-based processes.
- **Objectivity:** Ensure a fair and unbiased admission process by evaluating applicants based on predefined and objective criteria.